

GLUECKSBURG-MEIERWIK, Germany

WMO: 100330

Lat: 54.8272N

Lon: 9.5058E

Elev: 109

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	23.2	15.4	12.0	22.7	18.3	13.8	25.1	43.6	44.5	39.5	44.1	8.4	120	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	79.0	66.7	75.9	65.5	73.0	63.9	68.6	76.2	66.7	73.6	65.0	71.2	9.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.7	72.8	64.1	89.9	70.2	62.4	84.8	68.4	32.8	76.5	31.2	73.2	30.0	70.8	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.6	29.1	25.3	16.8	83.8	3.3	3.3	14.4	86.2	12.5	88.2	10.7	90.0	8.3	92.5		
(5)			16.0	71.8	3.3	1.5	13.7	72.9	11.8	73.8	9.9	74.6	7.6	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	35.2	35.8	39.2	46.3	53.5	58.8	63.3	63.4	57.9	50.5	42.7	37.6		
	DBStd	11.55	6.51	6.12	5.56	5.35	5.27	4.80	5.00	4.62	4.19	5.20	5.56	6.55		
	HDD50	2020	458	399	335	136	24	0	0	0	1	58	225	384		
	HDD65	6017	923	818	799	561	357	195	95	87	215	450	669	848		
	CDD50	1569	0	0	1	25	133	264	412	417	238	73	6	0		
	CDD65	95	0	0	0	0	1	9	43	39	3	0	0	0		
	CDH74	476	0	0	0	1	7	55	237	170	6	0	0	0		
	CDH80	52	0	0	0	0	0	5	26	21	0	0	0	0		
(14) Wind	WSAvg	12.0	13.8	13.7	13.3	11.2	11.2	11.8	10.9	10.4	11.1	12.0	12.2	13.0		
(15) Precipitation	PrecAvg	31.7	2.9	2.2	1.9	1.5	2.0	2.6	3.3	3.4	3.0	3.1	2.6	3.0		
	PrecMax	40.5	6.2	5.0	4.4	4.5	3.6	5.5	6.6	8.5	5.7	6.7	6.3	6.4		
	PrecMin	19.4	0.0	0.4	0.2	0.1	0.6	0.0	0.9	1.4	0.4	1.0	0.4	1.1		
	PrecStd	4.9	1.5	0.9	1.0	0.9	0.9	1.3	1.6	1.6	1.5	1.1	1.5	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	51.6	57.5	66.4	74.8	79.1	82.7	82.7	74.2	66.0	57.4	53.2		
		MCWB	48.2	45.9	49.3	55.0	62.8	65.7	68.2	67.6	65.1	60.3	54.7	50.8		
	2%	DB	47.7	48.1	52.2	62.5	70.2	75.1	79.9	78.6	70.4	62.3	54.7	50.2		
		MCWB	46.3	45.5	46.6	52.9	60.1	64.8	67.2	67.2	62.6	58.1	52.4	48.7		
	5%	DB	45.3	45.8	49.4	58.8	66.6	71.2	76.8	75.2	67.7	59.4	52.1	48.2		
		MCWB	44.0	44.1	45.3	51.0	57.3	61.9	65.9	65.5	60.9	56.2	50.0	46.9		
	10%	DB	43.9	44.2	46.7	55.4	63.3	67.8	73.3	71.7	64.7	57.5	50.2	46.3		
		MCWB	42.6	42.5	43.9	49.5	55.5	60.2	64.2	64.0	59.4	54.8	48.4	45.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.8	47.7	50.6	57.0	64.5	68.6	71.1	70.9	66.2	60.6	55.6	51.2		
		MCDB	49.5	49.5	54.7	64.4	72.5	76.9	79.3	78.1	71.7	64.1	57.4	52.6		
	2%	WB	46.0	45.7	47.9	54.2	61.2	65.6	69.0	68.4	63.7	58.7	52.8	48.9		
		MCDB	47.0	47.2	51.1	60.5	67.8	73.0	77.2	75.4	68.7	61.3	54.5	49.8		
	5%	WB	44.1	43.9	45.8	51.9	58.8	63.2	67.0	66.5	61.9	56.9	50.6	46.8		
		MCDB	45.0	45.3	48.2	57.4	65.0	69.1	74.2	73.0	66.1	59.2	52.0	47.7		
	10%	WB	42.4	42.3	44.2	49.9	56.5	61.1	65.0	64.9	60.2	55.3	48.6	44.9		
		MCDB	43.4	43.7	46.3	54.5	62.1	66.6	71.6	70.5	63.9	57.5	49.9	45.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	7.1	9.5	13.0	13.9	13.1	13.4	12.9	11.0	9.0	7.2	6.6		
		MCDBR	7.2	8.5	14.0	18.4	19.7	19.2	19.1	17.9	14.7	10.5	7.7	7.1		
		MCWBR	6.9	7.1	9.4	10.7	11.3	10.8	9.4	9.2	8.7	7.7	6.9	7.1		
	5% WB	MCDBR	6.9	7.7	12.0	16.4	17.7	16.8	17.2	15.8	12.7	9.8	7.5	7.0		
		MCWBR	6.8	6.8	8.9	10.5	11.2	10.4	9.5	9.1	8.4	7.7	7.3	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.341	0.368	0.400	0.393	0.389	0.406	0.405	0.382	0.359	0.337	0.310		
	taud		2.270	2.297	2.262	2.223	2.307	2.342	2.313	2.328	2.374	2.389	2.329	2.265		
	Ebn at Noon		190	232	252	260	270	272	264	257	246	223	184	167		
	Edh at Noon		19	25	33	39	38	37	37	35	29	23	18	15		
(44) All-Sky Solar Radiation	RadAvg		161	361	773	1292	1654	1725	1618	1308	923	490	200	117		
	RadStd		23	52	69	146	163	140	184	98	100	57	26	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	-232	N/A	N/A
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-266	N/A	N/A

Nomenclature: See separate page